

Chapter 13.

* Definition: Let A be an $n \times n$ matrix. We say that a number λ is an eigenvalue of A if there is a vector $v \neq 0$ such that

$$Av = \lambda v$$

such a vector is called an eigenvector of A corresponding to the eigenvalue λ .

* λ is an eigenvalue of $A \iff A - \lambda I$ is singular
 $\iff \det(A - \lambda I) = 0$

* Procedure of finding eigenvalues and eigenvectors of a matrix A .

1. Solve λ in $\det(A - \lambda I) = 0$ (characteristic equation).

2. Put λ found in step 1 in the equation.

$$(A - \lambda I)v = 0$$

and then solve v from the above singular system.

* A real-valued matrix can have complex eigenvalues, and the complex eigenvalues must come in conjugate pairs.

* Spectral radius of an $n \times n$ matrix M

$\rho(M)$ = maximum of the absolute values of the eigenvalues of M .

$\rho(M) \leq \|M\|$ where $\|M\|$ is the norm of M defined in terms of a vector norm $\|\cdot\|$.

* The iterative method $Sx^{new} = b + Tx^{old}$ converges if and only if $\rho(S^{-1}T) < 1$.

* $\prod_{j=1}^n \lambda_j = \lambda_1 \lambda_2 \dots \lambda_n$ where $\lambda_1, \dots, \lambda_n$ are eigenvalues of A_{nn} , then

$\sum_{j=1}^n \lambda_j = \lambda_1 + \lambda_2 + \dots + \lambda_n$

$\det(A) = \prod_{j=1}^n \lambda_j$

$\text{tr}(A) = \sum_{j=1}^n \lambda_j$

$\hookrightarrow$ trace of A , i.e., the sum of diagonal component of A .

Chapter 14.

* Definition. An eigenvalue of A , λ is dominant if $|\lambda|$ is strictly greater than the absolute values of all other eigenvalues.

* Power's method. — find the dominant eigenvalue of a matrix A .

1. Choose randomly a starting vector $x \neq 0$

2. Generate the approximations α to the dominant eigenvalue of A by applying the iteration formulas

$$z = \frac{x}{\|x\|_2}, \quad x = Az \quad \text{and} \quad \alpha = z^T x.$$

3. Stop when

$$\frac{|\alpha^{\text{old}} - \alpha^{\text{new}}|}{|\alpha^{\text{new}}|} < \text{prescribed tolerance } \epsilon.$$

* The Gerschgorin Circle Theorem

— used to give a rough estimate to the eigenvalue of a matrix A and determine whether A has a dominant eigenvalue.

* The inverse power method and The shifted inverse power method

Chapter 15.

* Definition. The vectors $v_1, v_2, \dots, v_n \in \mathbb{R}^m$ are linearly dependent if one of them is a linear combination of the others. They are linearly independent if they are not dependent.

* $v_1, v_2, \dots, v_n \in \mathbb{R}^m$ are linearly dependent

$\Leftrightarrow$ we can find n numbers $c_1, c_2, \dots, c_n$ which are not all zero such that

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0$$

$\Leftrightarrow \underline{[v_1 \ v_2 \ \dots \ v_n] X = 0}$ has non-trivial solution.
 $\hookrightarrow m \times n$ homogeneous system.

Remark: When $n=m$,

$v_1, v_2, \dots, v_n$ are linearly dependent (independent)

$\Leftrightarrow [v_1 \ v_2 \ \dots \ v_n]$ is singular (nonsingular)

* Orthogonal vectors.

Column- m -vectors x and y are orthogonal if $x^T y = 0$

A set of vectors $v_1, \dots, v_n \in \mathbb{R}^m$ is orthogonal if any two of them are orthogonal.

$$\{v_1, \dots, v_n \text{ are orthogonal}\} + \{v_j \neq 0 \text{ for } j=1, \dots, n\}$$

$$\Rightarrow \{v_1, \dots, v_n \text{ are linearly independent}\}.$$

* The Gram-Schmidt Orthogonalization process.

— produce a set of orthogonal vectors from a given set of linearly independent vectors $v_1, v_2, \dots, v_n$.

The G-S process.

1. Let $u_1 = V_1$

2. For $j=2$ to n , let u_j be defined by.

$$u_j = V_j - \left(\frac{V_j^T u_1}{u_1^T u_1} \right) u_1 - \left(\frac{V_j^T u_2}{u_2^T u_2} \right) u_2 - \dots - \left(\frac{V_j^T u_{j-1}}{u_{j-1}^T u_{j-1}} \right) u_{j-1}$$

* Definition: We say that the vectors $V_1, V_2, \dots, V_n \in \mathbb{R}^m$ are orthonormal if they are orthogonal and $\|V_1\|_2 = \|V_2\|_2 = \dots = \|V_n\|_2 = 1$.

Given orthogonal vector $\{V_1, V_2, \dots, V_n\}$ such that $V_1 \neq 0, V_2 \neq 0, \dots, V_n \neq 0$, then the vectors

$$\left\{ \frac{V_1}{\|V_1\|_2}, \frac{V_2}{\|V_2\|_2}, \dots, \frac{V_n}{\|V_n\|_2} \right\}$$

are orthonormal.